

HYDRAULIC MACHINES

Module 3

- A **fluid machine** is a device which converts the energy stored by a fluid into mechanical energy **or vice versa**.
- Machines using liquid (mainly water, for almost all practical purposes) are termed as hydraulic machines.
- Hydraulic Machines convert

Hydraulic energy → Mechanical energy eg :
Turbine

mechanical energy → Hydraulic energy eg: pumps

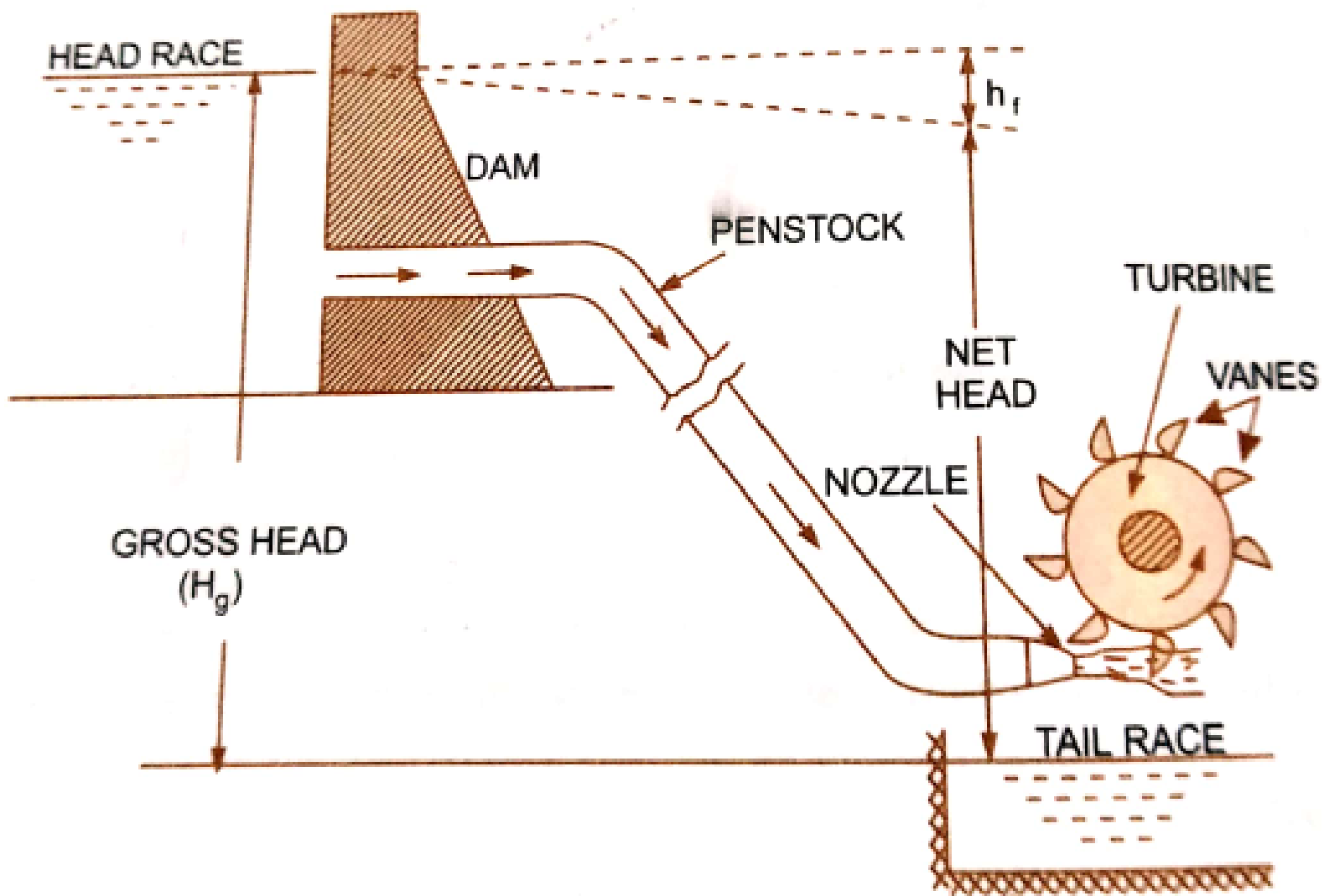
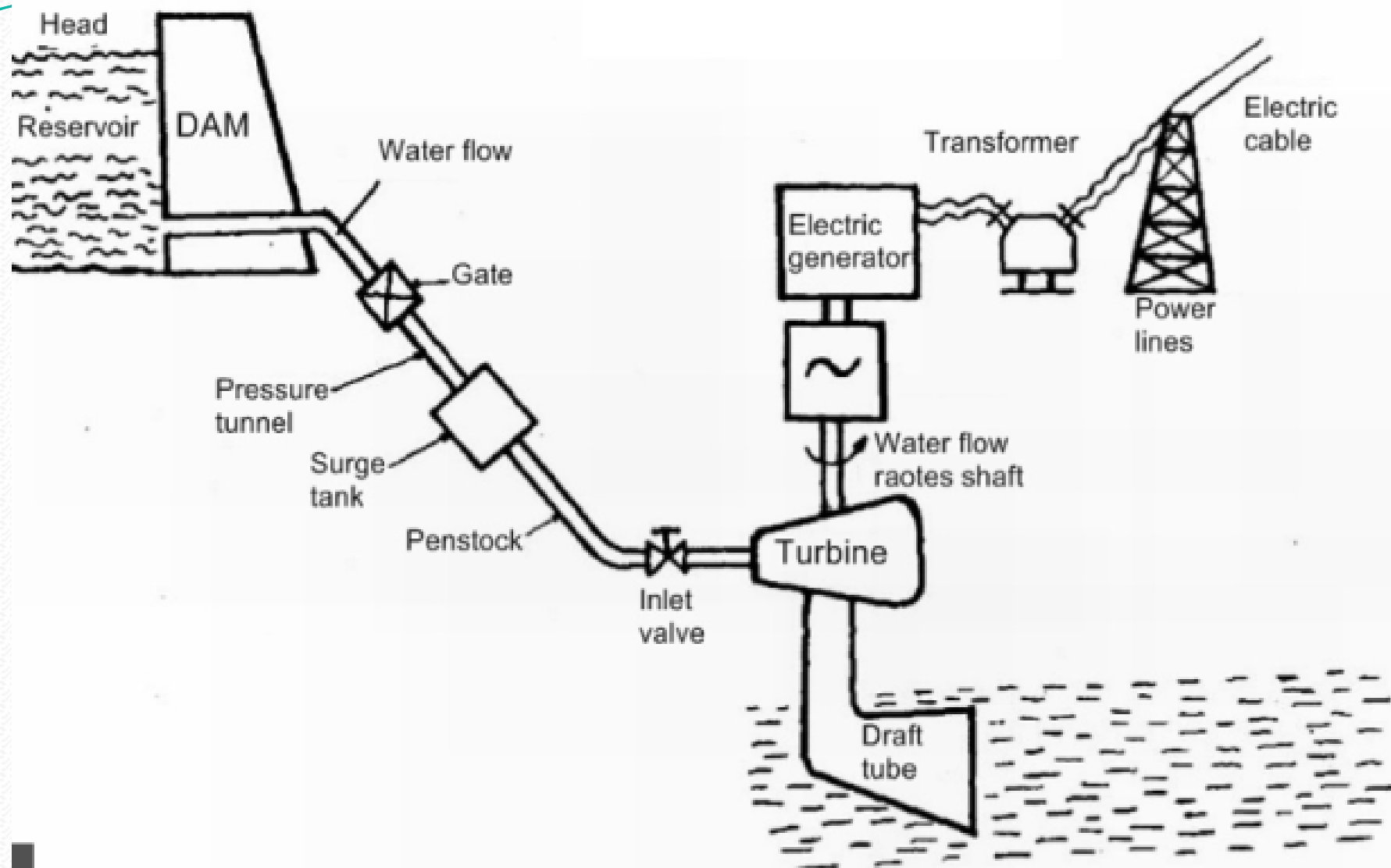


Fig. 18.1 *Layout of a hydroelectric power plant.*



Layout of hydro electric power plant

Classification of turbines

1. Based on type of energy at inlet

1. impulse turbine
2. Reaction turbine

2. Based on head and discharge

1. High head turbine (>250 m eg: Pelton turbine)
2. Medium head turbine (60– 250 m) eg: Francis turbine
3. Low head Turbine (Below 60m) eg: Kaplan turbine

3. Based on specific speed

1. Low specific speed (10 – 60) eg: pelton
- 2 . Medium sp speed (60 – 300) eg: francis
- 3 . High sp speed (300 – 1000) eg :
Kaplan

4. Based on direction of flow through runner

1. Radial Flow turbine eg : Francis turbine
- 2 . Axial flow eg : Kaplan
- 3 . Mixed flow eg : new Francis
- 4 . Tangential flow eg : Pelton wheel

Impulse turbine

- the energy available at inlet of turbine, is only kinetic energy.
- Before reaching the turbine the fluid's Pressure head is changed to velocity head by flowing through nozzle.

Eg Pelton wheel

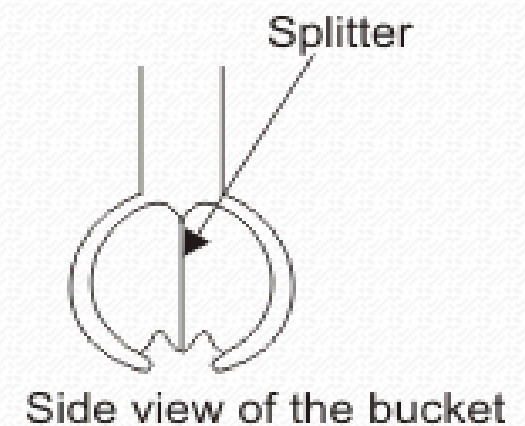
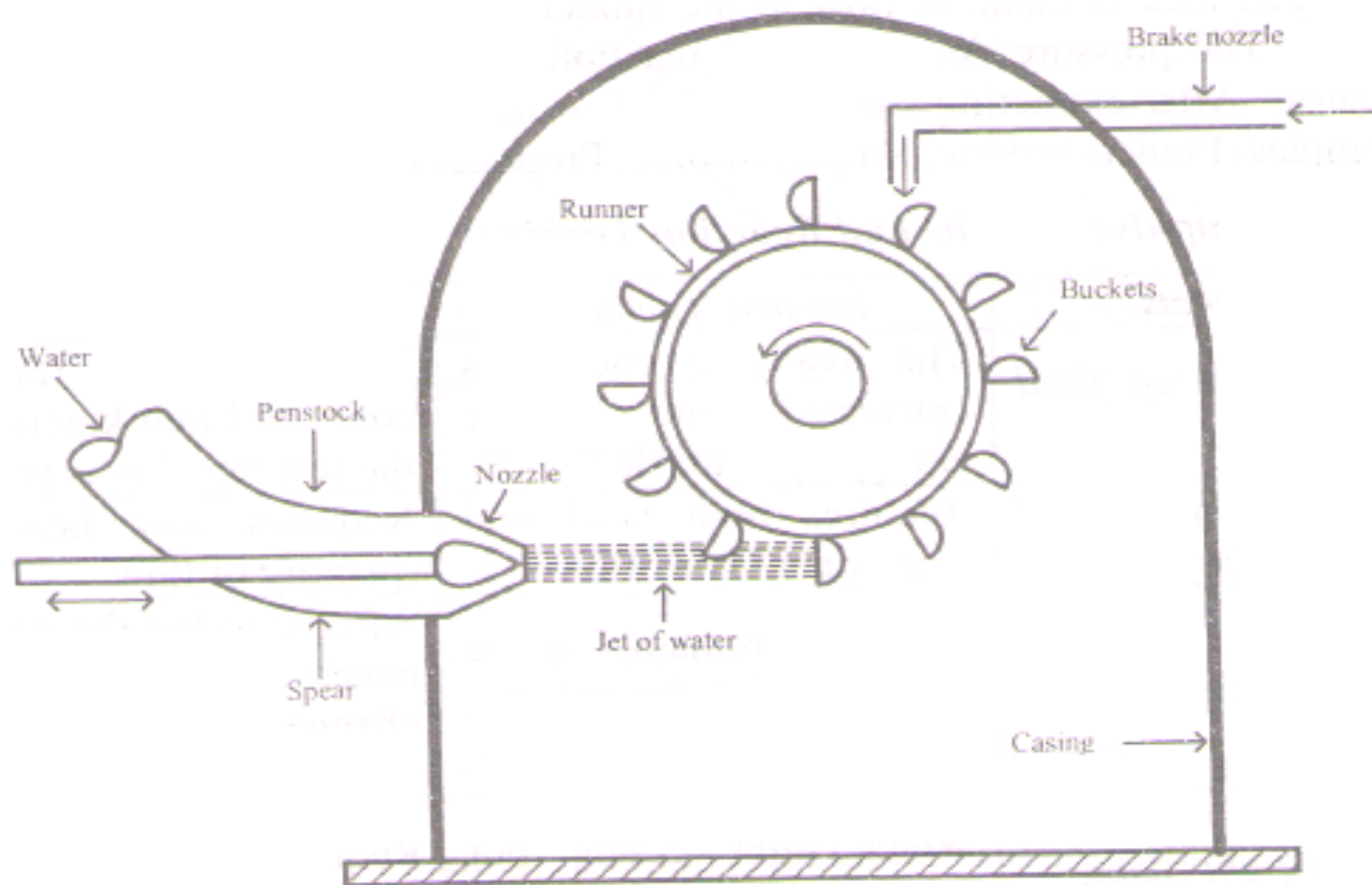
- Runner changes the direction of flow of a high velocity fluid jet.



Reaction turbine

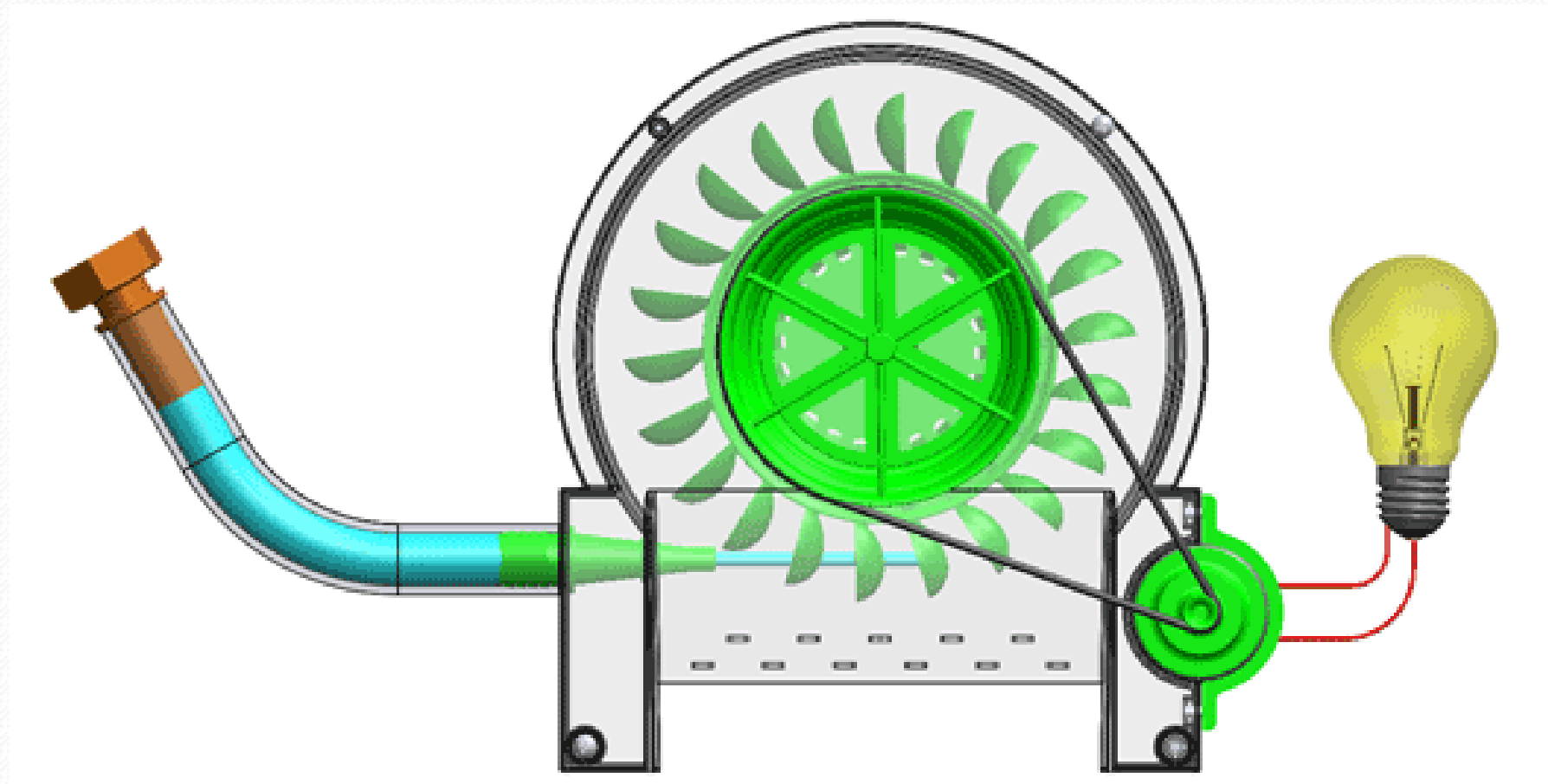
- Water at inlet possess KE as well as PE
- The pressure of the fluid reduces as it passes through the turbine runner blades, pressure energy changes to kinetic energy.
- This change in pressure energy is termed as reaction
- Requires water tight casing
- Eg: Francis Turbine, Kaplan Turbine etc.

Pelton turbine







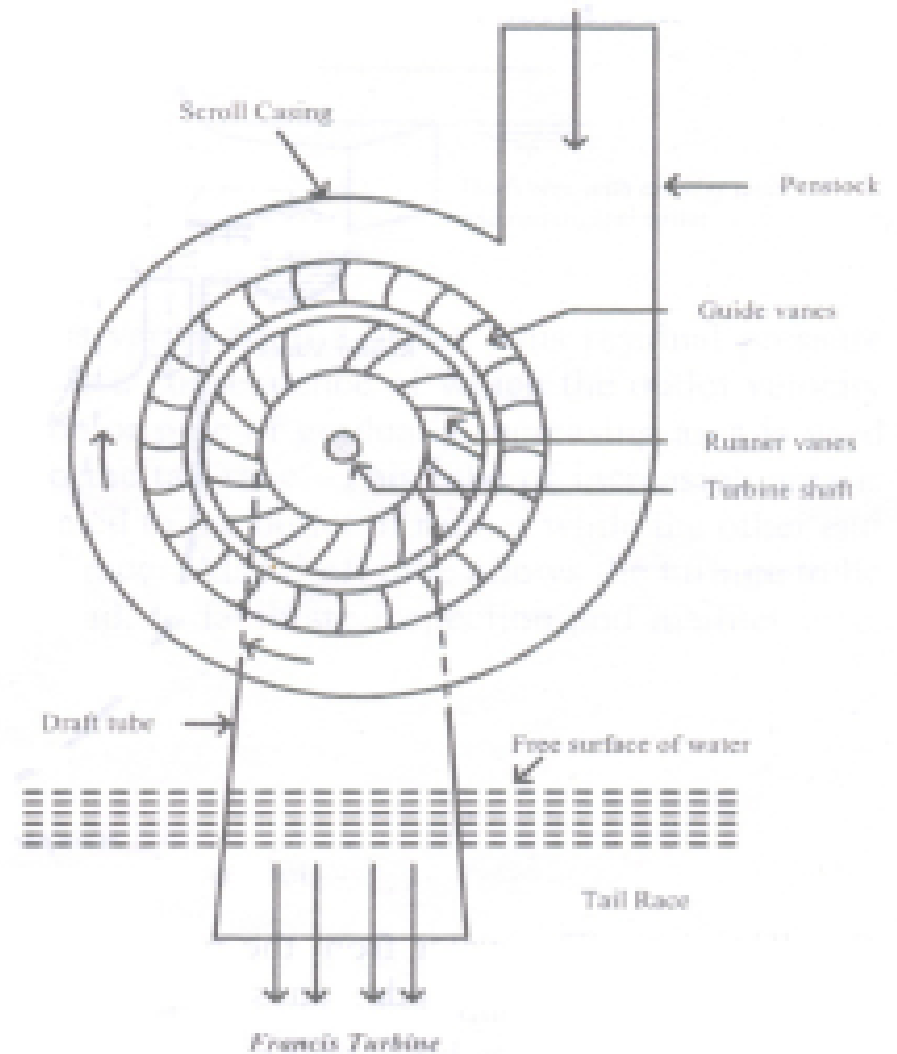
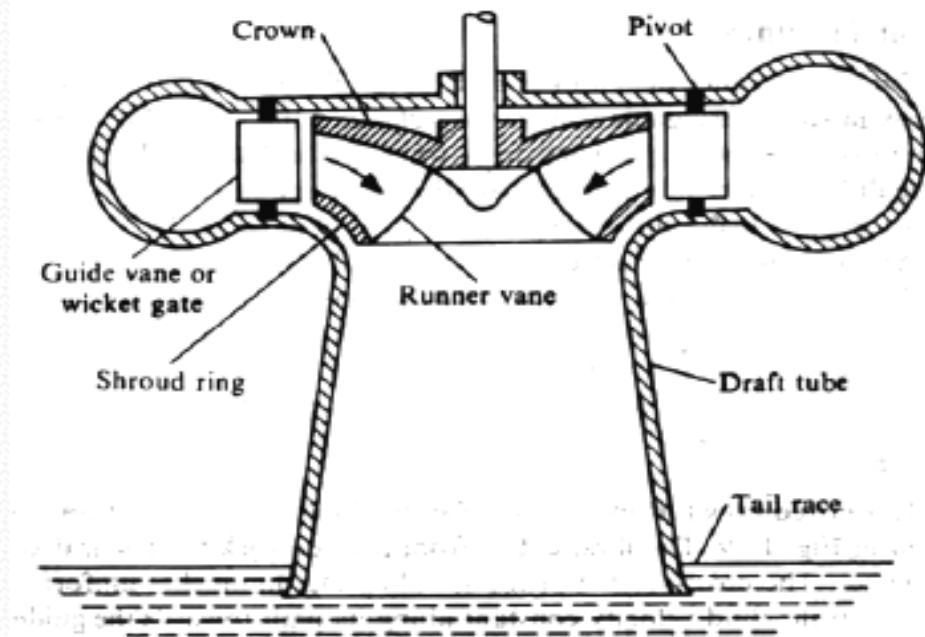


FRANCIS TURBINE

- The inward flow reaction turbine having radial discharge at outlet
- It is well suited for medium head operations such as head ranging from about 60 to 250 meters.
- Modern Francis turbine is a mixed flow reaction turbine
- The major parts of Francis turbine are

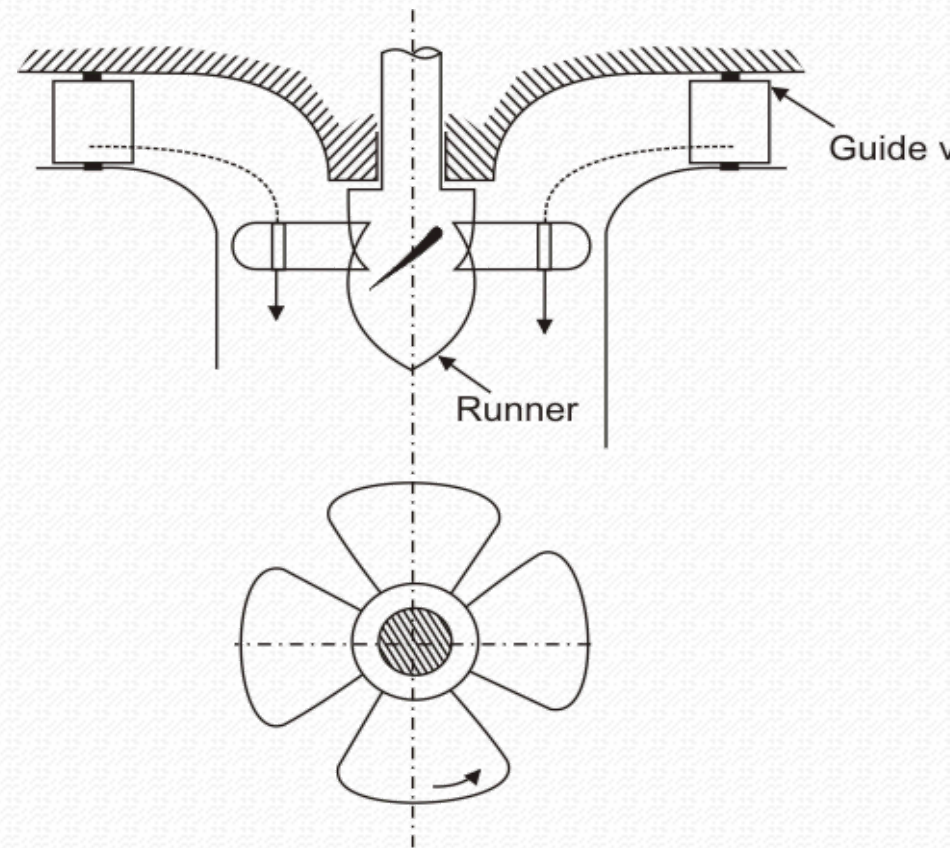
- ❖ Penstock,
- ❖ Spiral casing,
- ❖ Guide vanes,
- ❖ Runner
- ❖ Draft tube.

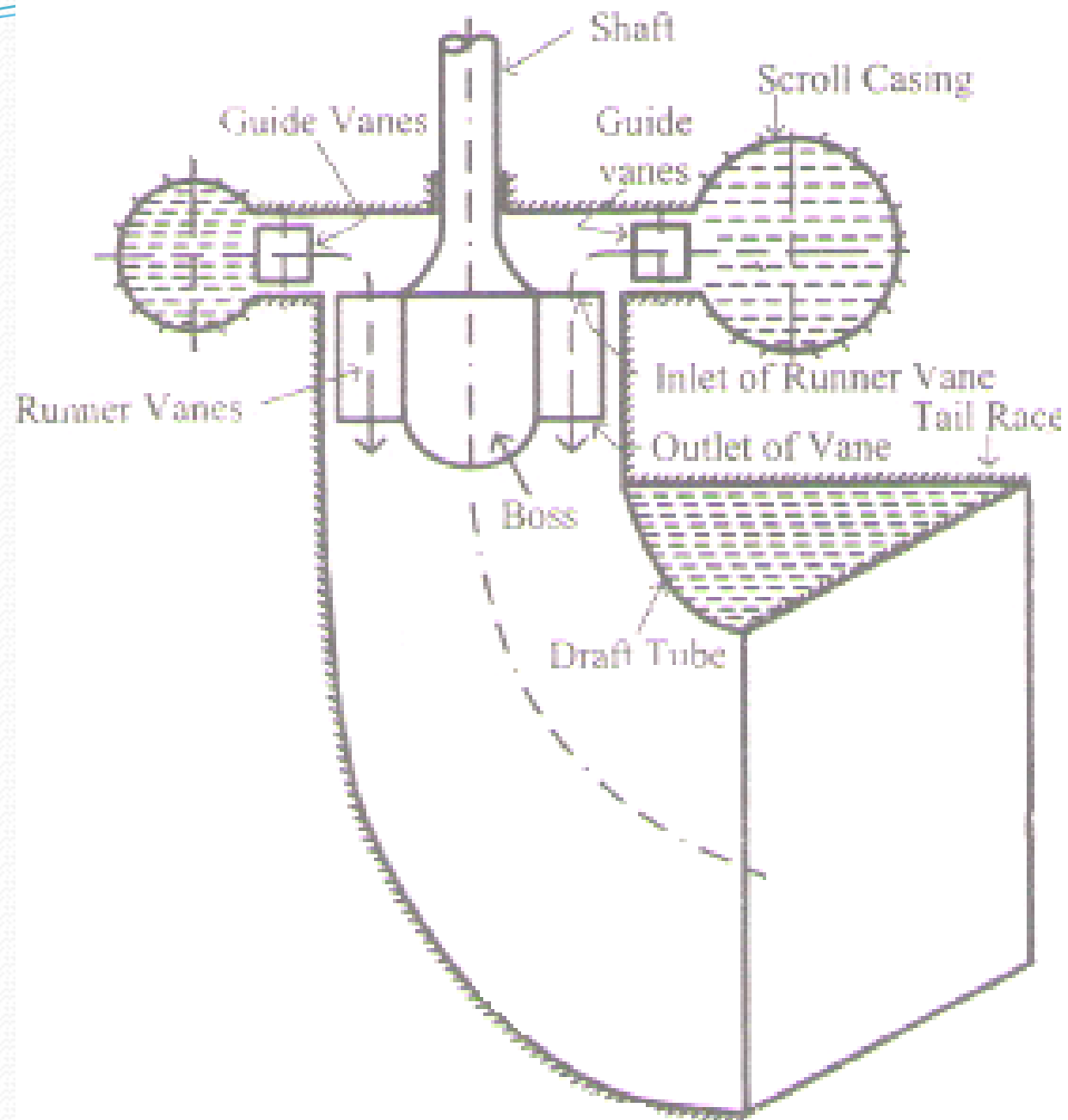
Francis Turbine



Kaplan turbine

- It is axial flow reaction turbine
- Low head reaction turbine
- Water enters to the runner axially and leaves axially
- Guide vanes and runner blades are adjustable





Draft Tube

It is a pipe of gradually increasing area which connects runner outlet to tailrace



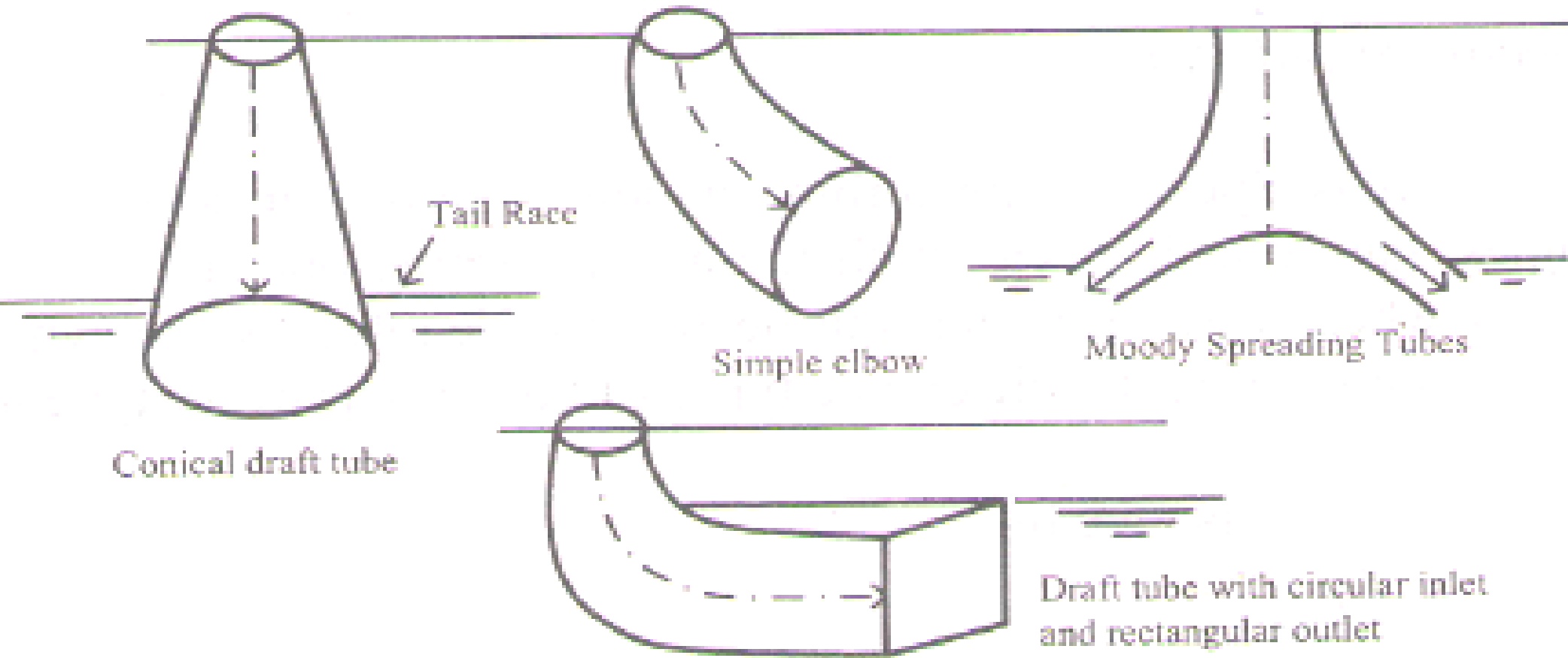
Turbine exit.

Less area, Velocity more, pressure less than atm



More area, Velocity less, more pressure

Different types of Draft Tube



Specific Speed (N_s)

- It is Speed of the turbine which is similar in shape, geometrical dimensions, blade angles, gate openings etc. with the actual turbine but of such a size that it will develop unit power working under unit head.
- It is denoted by the symbol N_s . The specific speed is used in comparing different types of turbines as every type has different specific speed.
- Specific Speed is give by the equa

$$N_s = \frac{N \sqrt{P}}{H^{5/4}}$$

P is Power Develop

H is Head under which Turbine is working

N is Speed of actual turbine

Pump

- Device which converts mech energy to hydraulic energy.
- Pump is a mechanical device which increases the pressure energy of a liquid.

classification

1. Positive displacement pumps

eg: Reciprocating pump, Gear pump

2. Rotodynamic pumps

eg: Centrifugal pump

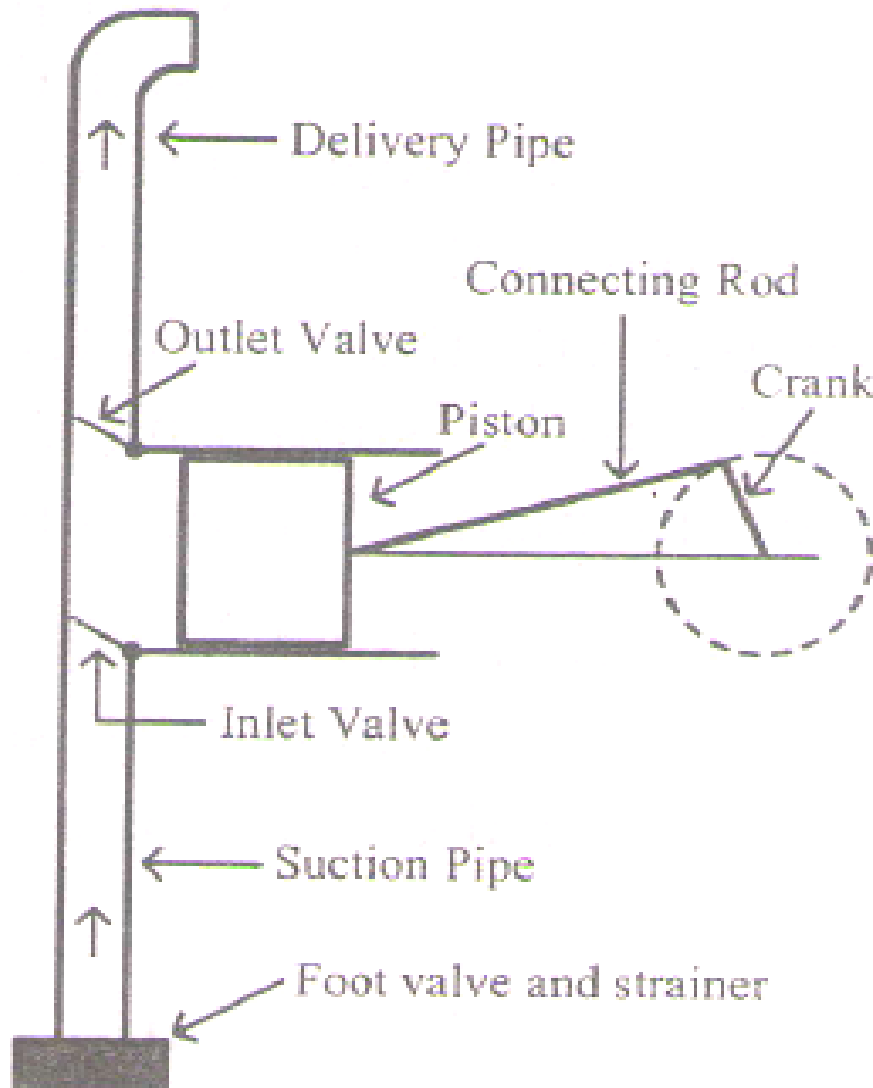
Positive displacement Pumps (Hydrostatic pumps)

- The positive displacement pump operates by alternating of filling a cavity and then displacing a given volume of liquid.
- Positive displacement pumps physically displace the fluid ; hence closing the outlet valve of a positive displacement pump will result in a continual build up in pressure resulting in mechanical failure of either pipeline or pump.

Rotodynamic pumps

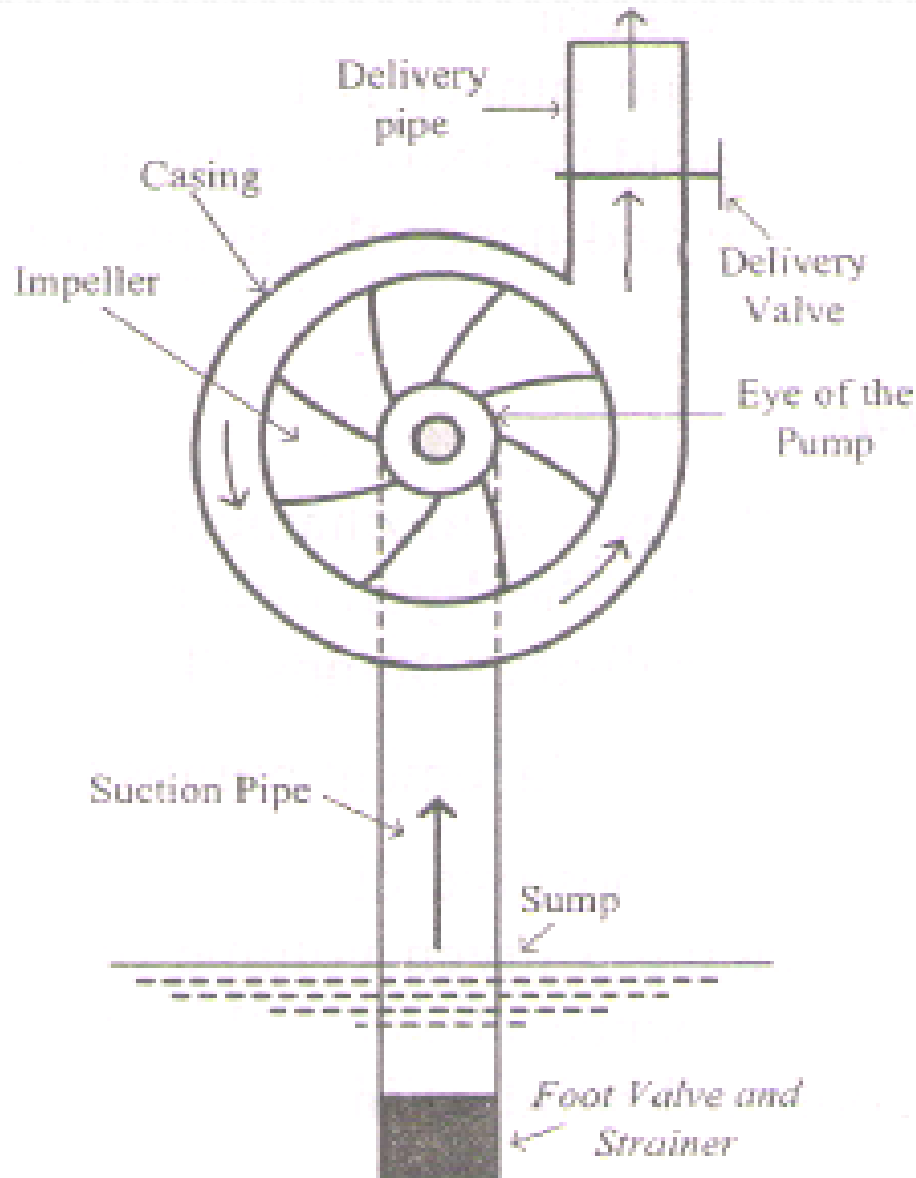
- The rotodynamic pump produces a head and a flow by increasing the velocity of the liquid through the machine with the help of a rotating vane impeller.
- Rotodynamic pumps can be safely operated under closed valve conditions (for short periods of time as heating of the fluid takes place) and hence there is no need for safety devices to prevent pump damage as air pressure developed automatically balances out on each side of the pump casing and stops the pumping.

Reciprocating Pump

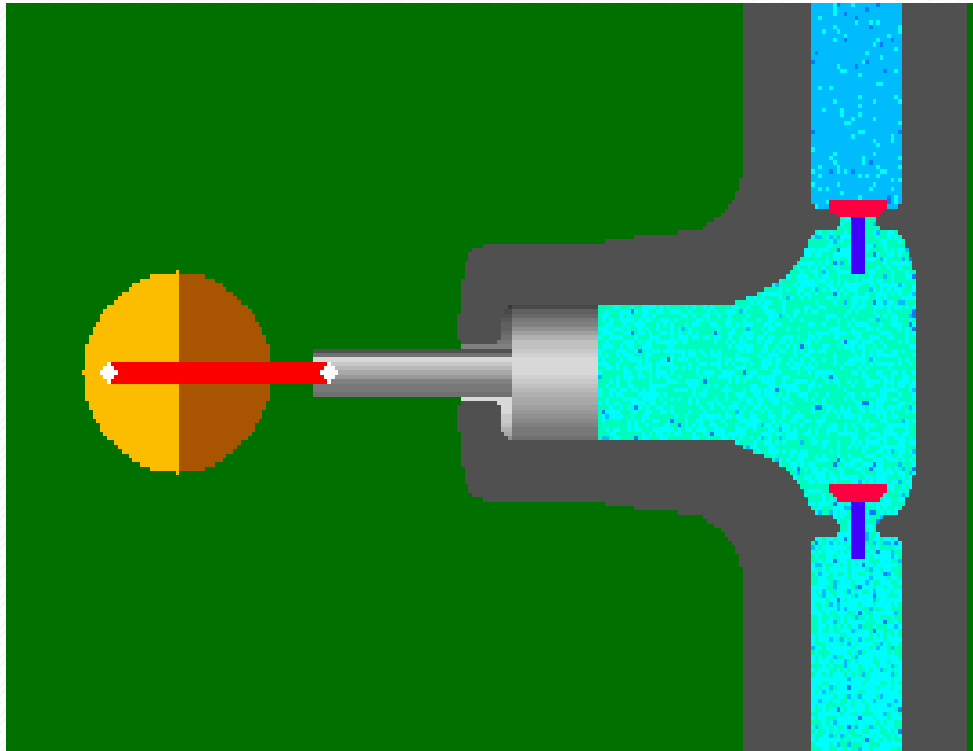


1. Foot valve & strainer.
2. Suction pipe.
3. Inlet valve.
4. Piston
5. Connecting rod & crank
6. Outlet valve.
7. delivery pipe

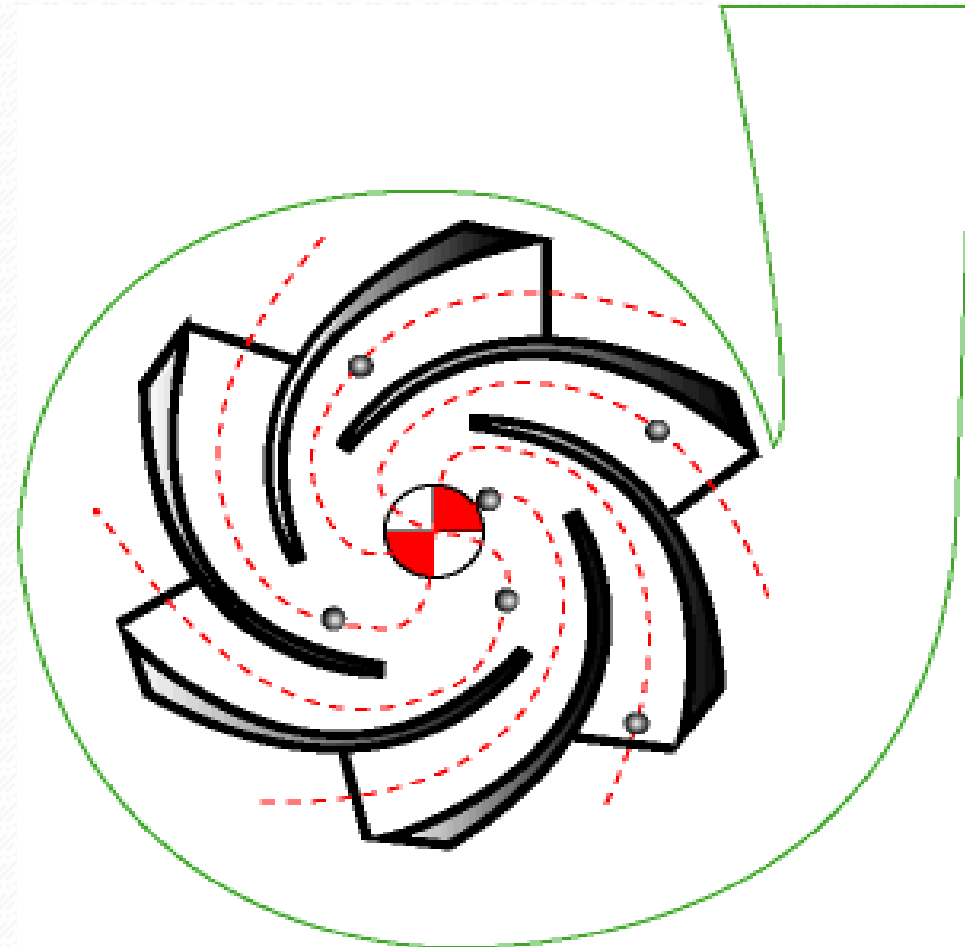
CENTRIFUGAL PUMP



1. Foot valve & strainer
2. Suction pipe
3. Impeller
4. Casing
5. Delivery pipe



Reciprocating pump



centrifugal pump

